BAJATO LIGHTS LLP

Product Requirements Document

Store Inventory Management Automation Tool

Document Version	v1.0
Status	Draft — For PM & Tech Lead Review
Date	May 2025
Department	Production & Stores
Business Unit	Manufacturing
Stakeholders	Store Team, Production Team, Purchase Team, Operations Manager
Platform	AppSheet (Google Sheets backend) + Email Alert Engine
Reference App	BLP-Inventory APP Sheet

1. Executive Summary

Bajato Lights LLP currently manages all raw material and consumable inventory via multiple formula-driven Google Sheets. This causes frequent data mismatches, delayed issuances, last-minute purchase alerts, and high dependency on experienced store staff. The Store Inventory Management Automation Tool will replace this workflow with a centralized, real-time AppSheet application backed by Google Sheets, providing role-based access for Store, Production, and Purchase teams.

This PRD defines the complete product vision, functional requirements, user stories, acceptance criteria, non-functional requirements, and annotated wireframes derived from the existing AppSheet interface screenshots for handoff to the design and development teams.

2. Product Vision & Goals

2.1 Vision Statement

To digitize and centralize inventory management across all store locations, enabling real-time stock visibility, digital material request workflows, proactive low-stock alerts, and SKU-based item identification — eliminating the inefficiencies and errors inherent in spreadsheet-based operations.

2.2 Business Objectives

- Provide 100% real-time stock visibility across all departments and locations.
- Reduce material issuance time by 70% through digital request and approval workflow.
- Enable proactive purchasing with a minimum 7-day advance low-stock alert.
- Reduce manual data-entry errors by more than 80%.
- Improve store staff productivity by 30–50%.
- Standardize all materials under unique SKU mapping for fast identification.

3. Problem Statement

3.1 Current Pain Points

- Store team maintains multiple Google Sheets with complex formulas — sheets frequently go out of sync.
- Incorrect or broken formulas produce wrong stock values without any error indication.
- Production team must manually check sheets to confirm material availability — no request tracking.
- Purchase team is only notified of shortages at the last minute, causing procurement delays.
- New employees cannot identify materials without guidance from experienced staff — no SKU map.
- No centralized real-time dashboard for any stakeholder.
- Manual stock updates are time-consuming and prone to omission.

3.2 Business Impact

- Production delays due to material unavailability or incorrect stock data.
- Increased operational risk from unplanned procurement.

- High dependency on individual employees who understand the sheet structure.

4. Success Metrics (KPIs)

Metric	Baseline (Current)	Target After Launch
Material issuance time	Manual — 15–30 min/request	70% faster (< 5 min)
Stock data accuracy	~60–70% (formula errors)	95%+
Purchase lead time	Last-minute (< 24–48 hrs)	Minimum 7-day advance alert
Manual data entry errors	High — no validation	Reduced by 80%+
Store staff productivity	Baseline	30–50% improvement
Item identification time	Depends on experienced staff	Near-zero with SKU search
Pending request visibility	None	Real-time dashboard for all roles

5. Scope

5.1 In Scope — Phase 1

- Item Master with SKU management and search.
- Stock Inward module (GRN entry with supplier details).
- Stock Outward / Issue module with real-time deduction.
- Digital Material Request workflow (Production → Stores approval).
- Low-stock alert engine (email + in-app notification).
- Stock dashboard and reporting (CSV/PDF export).
- Role-based user access: Store, Production, Purchase, Admin.
- Zero Initial Stock view — items with no current stock.
- Issued Items List with date-grouped history.
- Rejection List with vendor-return notification capability.
- Return Items List.
- Entry Count tracking.
- Catalogue / Product browser with search.

5.2 Out of Scope — Phase 2

- ERP integration (SAP / Tally / Odoo).
- Automatic purchase order generation.
- Barcode / QR scanning for material movement.
- Supplier portal for direct delivery confirmation.

6. User Personas & Roles

Role	Access Level	Key Actions	Alert Needs
Store Team (Admin/Store)	Full access	GRN entry, issue materials, manage item master, approve/reject requests, process returns and rejections.	Low-stock alerts, pending request notifications.
Production Team (Request)	Request & track only	Raise material requests with SKU + quantity + date, track approval status, view issued items.	Approval status updates.
Purchase Team	Read-only + alerts	View stock levels, low-stock dashboard, critical items list.	Low-stock email alerts with SKU, qty, and reorder level.
Admin	Full + configuration	Edit item master, set min-level thresholds, manage users, view all reports.	System-level alerts.

7. User Stories & Acceptance Criteria

US-01: Real-Time Stock Inward (GRN)

As a Store User, I want to log received materials against a GRN, so that inventory stock is updated immediately and accurately.

Acceptance Criteria

- System updates stock balance in real time upon GRN submission.
- Mandatory fields validated: Item SKU, Quantity Received, Date, Supplier.
- Incorrect or duplicate SKU entries display a clear validation error.
- Option to upload delivery challan or receipt document.
- GRN entry appears in Issued Items List history with timestamp.

US-02: Digital Material Request (Production)

As a Production User, I want to raise a digital material request, so that I can track approval status without calling the store team.

Acceptance Criteria

- Request form captures: SKU, Required Quantity, Required Date, Department.
- System generates a unique Request ID on submission.
- Store team receives an in-app notification and email for each new request.
- Production user can view real-time approval status: Pending / Approved / Partially Approved / Rejected.
- Request history is retained and searchable.

US-03: Material Issuance

As a Store User, I want to issue materials against an approved request, so that stock is deducted in real time and the production team is served efficiently.

Acceptance Criteria

- Store team cannot issue more than the available stock balance.
- Partial issuance is supported; remaining quantity stays as pending.
- Issued quantity is deducted from live stock immediately.
- Issue log captures: Issuer Name, Date, Time, Request ID.
- If stock is insufficient, the system shows an alert to the store team before issuance.

US-04: Low-Stock Alerts

As a Purchase User, I want to receive automatic low-stock alerts, so I can initiate procurement before stock runs out.

Acceptance Criteria

- Alert triggers only when current stock < minimum level threshold.
- Alert includes: SKU, Item Name, Current Quantity, Minimum Level, Reorder Quantity.
- Alert is sent via both email and in-app notification.
- Low-stock items appear in a 'Critical Items' section on the dashboard.
- Alert fires after every stock transaction, not just on a scheduled basis.

US-05: Rejection Processing

As a Store User, I want to log rejected items and notify the vendor, so that rejected material is tracked and returned efficiently.

Acceptance Criteria

- Rejection List displays: SKU, Item Name, Rejected Quantity, and send-to-vendor action.
- Vendor notification can be sent directly from the rejection list via the airplane/send icon.
- Rejected items do not count toward available stock.

US-06: Return Items

As a Store User, I want to log items returned from production, so that stock is updated correctly.

Acceptance Criteria

- Return Items List captures: SKU, Item Name, Quality/Material type, returned quantity.
- Returned quantity is added back to available stock in real time.
- Return history is searchable and filterable.

US-07: Zero Stock Visibility

As a Store or Purchase User, I want to see all items with zero stock at a glance, so I can prioritize procurement.

Acceptance Criteria

- 'Zero Initial Stock' view filters and displays all items with current stock = 0.
- Each zero-stock item shows SKU, Name, and action buttons.
- View updates in real time as stock is received or issued.

US-08: SKU-Based Product Search

As any user, I want to search for materials by SKU, name, or category, so I can find items quickly without needing expert knowledge.

Acceptance Criteria

- Global search bar is available on all main screens.
- Search works by SKU, Item Name, and Category.
- Results appear within 1 second of query entry.
- Catalogue view shows full item details: SKU, Category, Quality, Color, UOM, Current Stock, Image.

8. Functional Requirements

FR ID	Module	Requirement	Notes / Constraints
FR-01	Item Master	Store each item's: Unique SKU, Item Name, Material Type, Category, Quality, Color, UOM, Location, Min Level, Supplier (optional), Image.	Auto-validate SKU uniqueness on save.
FR-02	Item Master	Support SKU auto-generation with no duplicates allowed.	Format: [Category Code]-[Sequence], e.g. TL-301.
FR-03	Item Master	Global search by SKU, name, category across all screens.	Search must return results in < 1 second.
FR-04	GRN / Stock Inward	GRN form with mandatory fields: Item SKU, Qty Received, Date, Supplier.	Optional: upload delivery challan image/PDF.
FR-05	GRN / Stock Inward	Inward quantity adds to real-time Current Stock on submission.	Current Stock = Initial Stock + Inward - Outward + Returns.
FR-06	Stock Outward	Store selects an approved request and issues materials; system deducts qty from stock.	Cannot issue > available stock; partial issuance allowed.
FR-07	Stock Outward	Issue log records: Issuer Name, Date, Time, Request ID.	Log is immutable once saved.
FR-08	Material Request	Production raises request with: SKU, Required Qty, Required Date, Department.	System generates unique Request ID (auto-increment).
FR-09	Material Request	Store team receives notification (in-app + email) on new request.	Notification includes Request ID, SKU, Qty, Required Date.
FR-10	Material Request	Store team can: Approve, Partially Approve, or Reject request.	Rejection requires a reason field.
FR-11	Material Request	Full request history retained and searchable by Request ID, SKU, date.	Requestor can see status in real time.
FR-12	Low-Stock Alert	After every stock transaction, system compares Current Stock vs Min Level.	Alert fires if Current Stock < Min Level.
FR-13	Low-Stock Alert	Alert sent to Store Team + Purchase Team via email and in-app notification.	Alert content: SKU, Item Name, Current Qty, Min Level, Reorder Qty.
FR-14	Low-Stock Alert	Critical Items section in dashboard lists all sub-threshold items.	Sorted by most critical (lowest stock %) first.

FR-15	Rejection List	Display all rejected items with: SKU, Name, Qty, and vendor-notify action.	Clicking send icon dispatches vendor notification.
FR-16	Return Items	Log returned materials; add qty back to Current Stock.	Captured fields: SKU, Qty, Quality, Date.
FR-17	Zero Initial Stock	Dedicated view showing all items where Current Stock = 0.	Updates in real time; shows action buttons for each item.
FR-18	Dashboard	Store dashboard shows: current stock by SKU/location/category, consumption trends, critical items, pending requests, stock-in vs out summary.	Filters: location, date range, SKU.
FR-19	Dashboard	All reports exportable as CSV and PDF.	Export button visible on dashboard and all list views.
FR-20	User Access	Three roles: Store (full), Production (request & track), Purchase (read-only + alerts).	Admin role can edit item master and thresholds.
FR-21	Entry Count	Track total GRN entries and issuance entries per day/week/month.	Visible in Entry Count screen for admin/store users.
FR-22	Catalogue	Product browser showing all items with image, SKU, stock level, category.	Allows quick inward/request-item initiation from catalogue view.
FR-23	Google Sheets Sync	All app data syncs to the backing Google Sheet in real time.	BLP-Inventory APP Sheet remains the data source of truth.

9. Non-Functional Requirements

9.1 Performance

- All screen transitions must complete within 2 seconds on a standard 4G connection.
- Search results must render within 1 second of query entry.
- Stock balance updates must reflect within 3 seconds of GRN or issue submission.

9.2 Reliability & Data Integrity

- Current Stock formula must be enforced at the database level: Initial Stock + Inward − Outward + Returns − Rejections.
- No transaction can result in negative stock; system must block and alert.
- All transactions are logged with user identity, timestamp, and device.

9.3 Usability

- Interface must be operable by users with no prior technical training within one training session.
- Store and Production users have differentiated interfaces — requestor sees only request-relevant actions.

- Interface must be clean, minimal, and light-background as specified in the AppSheet wireframe.

9.4 Security & Access

- Role-based access enforced at both UI and data layer.
- Production users cannot access GRN or issuance forms.
- Purchase team has read-only access to stock data; cannot modify any records.
- All data stored in Google Workspace with organization-level access controls.

9.5 Compatibility

- AppSheet app must work on Android, iOS, and desktop web browsers.
- Google Sheets backend must be accessible to authorized Google accounts only.

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10. Annotated Wireframes

The following wireframes are derived from the BLP-Inventory AppSheet interface screenshots. Each wireframe shows the screen layout and key annotations for the design and development teams.

10.1 Navigation Sidebar (All Screens)

```
                    [ LEFT NAVIGATION SIDEBAR ]
≡   [BLP-Inventory]   ·   [Search Bar — top of every screen]   [□] [□]
──────────────────────────────────────────────────────────────────────
□  Products            ← Default landing screen (Catalogue view)
□   Catalogue           ← Full item browser with image + stock
□  Requested Items     ← Production: request list | Store: approval
⊙   Zero Initial Stock  ← Items where Current Stock = 0
↪   Issued Items List   ← Date-grouped issuance history
□  Rejection List      ← Rejected items + vendor notify action
↩   Return Items List   ← Returned material log
□  Entry Count         ← Admin: daily/weekly entry totals
ℹ   About
□  Feedback
⊞   App Gallery
```

Annotation: Every option in the left panel has a differentiated view for Store users vs. Requestor (Production) users based on their assigned role. Store users see action buttons (Inward, Request Item, Return); requestors see only browse and request options. Interface is clean with white/light background per design specification.

10.2 Products / Catalogue Screen

```
                    [ PRODUCTS — STORE USER VIEW ]
```

```
[≡]  BLP-Inventory APP Sheet    [Search Products...]        [□] [□]
──────────────────────────────────────────────────────────────────────
Products
┌──────────────────────────────┐  Detail Panel (right on desktop)
│ AX-577                    ↑  │  ┌──────────────────────────────┐
│ ──────────────────────────   │  │ Item Name:  TARMINAL PATTI   │
│ TARMINAL PATTI           BM-83│  │ Stock:      1,811            │
│                           ↑↓ │  │ Item SKU:   DAG-SMP-Q005-C13│
│ AX-467                       │  │ Cat. No.:   AX-577           │
│ KNOB REFLEX REFLECTOR     0  │  │ Category:   SHEET METAL      │
│                           ↑↓ │  │ Quality:    BRASS            │
│ AX-480                       │  │ Color:      COPPER           │
│ DASH BOARD INDICATOR LENS 400│  │ Current Stock: 0             │
│                           ↑↓ │  │ [Inward] [Request Item] [Return]│
│ DASH BOARD INDICATOR HSG  120│  └──────────────────────────────┘
└──────────────────────────────┘
```

Annotation: Left panel is scrollable product list (SKU + Name + current stock count). Tapping an item opens a detail panel on the right (desktop) or full screen (mobile). Store users see three action buttons: Inward, Request Item (for production-initiated inward), and Return. Production users see only 'Request Item'. Stock value highlighted — 0 shown in red.

10.3 Google Sheet — Current Stock Tab

```
                          [ GOOGLE SHEET: BLP-Inventory APP Sheet — Current Stock Ta
Col:  A        B              C           D    E          F          G      H      I       J
      Unique   Item SKU       Item Name   Cat  Category   Quality    Color  UOM    Inward  Out
Img
      No.                                 No.
 ──────────────────────────────────────────────────────────────────────────────────────────
0001  DA8-SMP  Q005-C10-V     CHASSIS     TL-301 SHEET   ZINC PLATED WHITE  Pcs    670     650
0002  DA8-SMP  Q005-C10-V     BODY        TL-303 SHEET   BLACK       WHITE  Pcs    450     0
0003  DA8-SMP  Q005-C10-V     CHASSIS     TL-302 SHEET   ZINC PLATED WHITE  Pcs    0       200
...
Tabs: Instructions | Current Stock | Item SKU MSt | Inward | Request Form | Return | Rejection se
```

Annotation: The Google Sheet is the data backbone of the AppSheet app. Current Stock column (M) = Initial Stock (L) + Inward (H) − Outward (I) + Return (J). Rows highlighted in red/orange indicate low or zero stock. Each tab maps to an AppSheet view: Current Stock → Products screen; Inward → GRN entry; Request Form → Requested Items; Return → Return Items List.

10.4 Google Sheet — User Access Tab

```
[ GOOGLE SHEET: User Tab — Role & Email Mapping ]

Col A          Col B       Col C
User Name      Role        Mail Id
──────────────────────────────────────────────────────────────
Store          Store       se@bajato.com
SFG            Request     a2@bajato.com
Pankaj Sir     Request     vp@bajato.com
Rajat Bajaj    Request     pe@bajato.com
Rahul          Request     pk@bajato.com
Admin          Store       im@bajato.com
Vishal         Request     sng@bajato.com
Lalit          Store       se@bajato.com
Manoj          Request     as@bajato.com
──────────────────────────────────────────────────────────────
Role = Store   → Full access + action buttons
Role = Request → Request & track only; no Inward/Return buttons
```

Annotation: The User tab in the Google Sheet drives AppSheet's role-based access. Each user's email is matched to a Role (Store or Request). Admin users get Store role plus item master edit access. Purchase team users (read-only) are to be added as a third role in Phase 1.

10.5 Requested Items Screen

```
[ REQUESTED ITEMS — STORE USER VIEW (approval interface) ]

[≡]  BLP-Inventory APP Sheet   [Search Requested Items...]   [✓] [□]
──────────────────────────────────────────────────────────────
Requested Items                                   [Filter ▼]
┌──────────────────────────────────────────────────────────┐
│ AX-701                                        420  ↑↓ ✎  │
│ AUX. LAMP LENS                                           │
├──────────────────────────────────────────────────────────┤
│ AX-701                                        800  ↑↓ ✎  │
│ AUX. LAMP BASE                                           │
├──────────────────────────────────────────────────────────┤
│ LG-004                                        115  ↑↓ ✎  │
│ 4x13 STEEL SCREW                                         │
├──────────────────────────────────────────────────────────┤
│ LG-031                                        400  ↑↓ ✎  │
│ 4X15 STEEL SCREW                                         │
└──────────────────────────────────────────────────────────┘
```

```
Store actions per row: [↑ Issue] [↓ Partial Issue] [✎ Edit/Reject]
```

Annotation: Store users see all pending requests with SKU, Name, and requested quantity. Action icons allow: full issue, partial issue, or edit/reject. Production users see the same list but only their own requests, with status badges (Pending / Approved / Rejected). Filter allows by date, department, and status.

10.6 Rejection List Screen

```
[ REJECTION LIST — STORE USER VIEW ]
[≡]  BLP-Inventory APP Sheet   [Search Rejection List...]    [□] [□]
Rejection List                                          [Filter ▼]
┌──────────────────────────────────────────────────────────────┐
│ TL-997   POLYBAG 10X10 FOR PACKING 150 GAUGE      31,844 ✎ ✈ │
├──────────────────────────────────────────────────────────────┤
│ TL-997   POLYBAG 7X15 FOR KIT 125 GEJ             20,000 ✎ ✈ │
├──────────────────────────────────────────────────────────────┤
│ TL-999   POLYBAG 6X10 FOR PACKING 150 GAUGE       14,877 ✎ ✈ │
├──────────────────────────────────────────────────────────────┤
│ SL-515   SIDE LAMP LENS                            1,200 ✎ ✈ │
├──────────────────────────────────────────────────────────────┤
│ TL-317   BODY GROMMET                              1,000 ✎ ✈ │
└──────────────────────────────────────────────────────────────┘
✈ = Send rejection notification directly to vendor from this screen
```

Annotation: Rejection List provides a summary of all rejected items with quantities. The airplane icon (✈) allows the store team to send a rejection notification directly to the vendor without leaving the app. Edit icon (✎) allows updating rejection details. Rejected quantities are excluded from Current Stock.

10.7 Issued Items List Screen

```
[ ISSUED ITEMS LIST — DATE-GROUPED HISTORY ]
[≡]  BLP-Inventory APP Sheet   [Search Issued Items List...]  [□] [□]
Issued Items List                                       [Filter ▼]
┌──────────────────────────────────────────────────────────────┐
│ 5/4/2026  ──────────────────────────────────────────────────│
│   TL-997  HADA                                          2    │
├──────────────────────────────────────────────────────────────┤
│ 5/3/2026  ──────────────────────────────────────────────────│
│   TL-999  BAJATO LOGO STICKER                         500    │
```

```
│   TL-997  T-GELO TAPE                                  1      │
│   LG-005  SMD PCB, 12V                                 420    │
├──────────────────────────────────────────────────────────────┤
│ 5/1/2026 ──────────────────────────────────────────────────── │
│   LG-902  8 x 6.5 SCREW PHILIP HEAD                   1605   │
└──────────────────────────────────────────────────────────────┘
Each row: [SKU] [Item Name] [Qty Issued] [Edit icon]
```

Annotation: Issued Items List is grouped by date (newest first) for easy audit. Clicking any entry opens the full issuance detail (Issuer Name, Request ID, Time). Edit icon allows admin to annotate but not change issued quantity (immutable log). Exportable as CSV/PDF.

10.8 Zero Initial Stock Screen

```
[ ZERO INITIAL STOCK — FILTER VIEW ]
[≡]  BLP-Inventory APP Sheet   [Search Zero Initial Stock...]  [□] [□]
──────────────────────────────────────────────────────────────────────
Zero Initial Stock                                              0 ▲
┌──────────────────────────────────────────────────────────────┐
│ LG-004                                                 0  ↑ ✎□ ✓│
├──────────────────────────────────────────────────────────────┤
│ LG-948                                                 0  ↑ ✎□ ✓│
├──────────────────────────────────────────────────────────────┤
│ LG-968                                                 0  ↑ ✎□ ✓│
├──────────────────────────────────────────────────────────────┤
│ HL-429                                                 0  ↑ ✎□ ✓│
├──────────────────────────────────────────────────────────────┤
│ HL-992                                                 0  ↑ ✎□ ✓│
└──────────────────────────────────────────────────────────────┘
Actions: [↑ Inward] [✎ Edit] [□ Link to supplier] [✓ Verify]
```

Annotation: This view is a critical procurement aid — all items with zero stock are surfaced immediately. Store team can initiate a GRN (Inward) directly from this screen. Purchase team uses this view to prioritize urgent procurement. The count badge (0 in above example) shows total zero-stock items.

11. Information Architecture & Navigation Flow

```
[ APP NAVIGATION FLOW ]
AppSheet Login (Google Account SSO)
      |
```

```
├─▶ [Store Role]  ──────────────────────────────────────────
│     ├─ Products (Catalogue)  → Item Detail → Inward / Return
│     ├─ Requested Items       → Approve / Partial / Reject
│     ├─ Zero Initial Stock    → Initiate Inward
│     ├─ Issued Items List     → Audit Log
│     ├─ Rejection List        → Vendor Notify
│     ├─ Return Items List     → Stock Update
│     ├─ Entry Count           → Admin Stats
│     └─ Catalogue             → Search + Browse
│
├─▶ [Request / Production Role] ──────────────────────────
│     ├─ Products (Catalogue)  → Item Detail → Request Item
│     ├─ Requested Items       → MY Requests + Status
│     ├─ Issued Items List     → MY Issued History
│     └─ Catalogue             → Browse Only
│
└─▶ [Purchase Role — Read Only] ──────────────────────────
      ├─ Products (Catalogue)  → View Only
      ├─ Zero Initial Stock    → View Only (procurement input)
      ├─ Critical Items Dashboard → Low-stock alerts
      └─ Reports               → Export CSV / PDF
```

12. Core Data Model (Google Sheet Tabs)

Sheet Tab	Purpose	Key Columns	AppSheet View
Current Stock	Master stock ledger	Unique No., Item SKU, Item Name, Cat No., Category, Quality, Color, UOM, Inward, Outward, Return, Initial Stock, Current Stock, Image	Products / Catalogue
Item SKU MSt	Item master reference	SKU, Name, Category, Min Level, Supplier, UOM	Item Master admin screen
Inward	GRN entries log	SKU, Qty Received, Date, Supplier, Challan Ref, Entered By	GRN Entry form
Request Form	Material request log	Request ID, SKU, Qty Requested, Required Date, Department, Status, Approved Qty	Requested Items screen
Return	Return material log	SKU, Qty Returned, Quality, Date, Returned By	Return Items List
Rejection sent	Rejected items log	SKU, Qty Rejected, Date, Vendor Notified (Y/N)	Rejection List

Rejection Status	Vendor response tracking	SKU, Rejection Date, Vendor Response, Resolution	Admin view
Category	Category master	Category Code, Category Name, Sub-category	Filter dropdowns
BOM	Bill of Materials	Product, Component SKU, Qty per Unit	Future: production planning
User	Role-based access	User Name, Role (Store/Request), Mail Id	App login & access control
Counts	Entry audit counters	Date, Inward Count, Outward Count, Total	Entry Count screen

13. Low-Stock Alert Engine — Logic Flow

```
                    [ ALERT ENGINE LOGIC ]
TRIGGER: Any stock transaction (GRN / Issue / Return / Rejection)
      |
      ▼
CHECK: Current Stock  vs.  Minimum Level Threshold (Item Master)
      |
      ├─ If Current Stock >= Min Level  →  No action
      |
      └─ If Current Stock < Min Level   →  ALERT SEQUENCE:
              ├─ 1. In-app notification → Store Team
              ├─ 2. In-app notification → Purchase Team
              ├─ 3. Email alert         → Store Team mail(s)
              ├─ 4. Email alert         → Purchase Team mail(s)
              └─ 5. Item added to 'Critical Items' dashboard section
                      Alert payload includes:
                        - SKU + Item Name
                        - Current Stock (units)
                        - Minimum Level Threshold
                        - Suggested Reorder Quantity
                        - Timestamp of trigger
```

14. Dependencies & Assumptions

14.1 Dependencies

- Google Workspace subscription (Sheets, Drive, Gmail) required for all users.
- AppSheet Pro or Enterprise license for role-based access and automation features.

- All store users must have organizational Google accounts.
- Item Master (SKU MSt tab) must be populated before go-live.
- Minimum level thresholds must be configured per item before alert engine activates.

14.2 Assumptions

- Initial stock data will be migrated from existing Google Sheets before launch.
- AppSheet will continue to use the existing BLP-Inventory APP Sheet as backend.
- Email alerts will be sent via AppSheet's native automation (no third-party email service required).
- Barcode scanning is out of scope for Phase 1; SKU entry is manual.
- BOM (Bill of Materials) tab is present in the sheet but out of scope for Phase 1 functional requirements.

15. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Initial stock data migration has errors.	Medium	High	Run parallel operations for 2 weeks post-launch; reconcile weekly.
Users revert to Google Sheets after launch.	Medium	High	Enforce access restrictions; provide training; monitor sheet edits.
AppSheet automation limits hit for large teams.	Low	Medium	Review AppSheet plan limits; upgrade if needed before go-live.
Min-level thresholds not configured, alerts don't fire.	High	High	Make threshold a mandatory field in Item Master; PM to verify before launch.
Email alerts go to spam.	Medium	Medium	Whitelist AppSheet domain; use org email addresses; test before go-live.
Role mis-assignment gives wrong access.	Low	High	User tab must be verified by admin before app is shared; role change process documented.

16. Indicative Timeline

Week 1	PRD sign-off | Item Master data cleanup and SKU validation
Week 2	AppSheet app configuration: views, forms, role-based access
Week 3	Alert engine setup (automation rules) | GRN & issuance form testing
Week 4	UAT with Store Team and Production Team (parallel run with Sheets)

Week 5	Bug fixes | User training sessions | Threshold configuration
Week 6	Go-live | Sheet write-access restricted | Monitor for 2 weeks
Phase 2 (TBD)	ERP integration | Barcode scanning | Auto PO generation

17. Open Questions for Stakeholders

- What is the exact formula for Current Stock — does it include rejection quantities as deduction?
- Should Purchase Team users be able to add comments to low-stock alerts?
- Is partial approval of a material request reflected back to the Production user with a revised quantity?
- How many locations/stores need separate stock tracking in Phase 1?
- Should rejected items be returned to stock, or are they always written off?
- Is Entry Count a simple transaction counter, or does it need to track by user as well?
- Should the Catalogue / Products screen be accessible without login for read-only browsing?

Sign-Off

This document requires review and approval from the following stakeholders before development begins:

Stakeholder / Role	Name	Approval & Date
Product Manager		
Operations Manager		
Store Team Lead		
Production Team Lead		
Tech / AppSheet Developer		

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